Interlock Sewing Machine



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: High speed interlock machine

What it is used for

To sew and finish fabric seams using a high-speed interlock sewing machine. To produce consistent stitches while maintaining correct thread tension, fabric feeding, and operator safety.

Main parts

- Machine head and needle bar
- Needles and looper assembly
- Thread guides and tension controls
- Presser foot and feed mechanism
- Motor, speed control, and foot pedal
- Lubrication system and safety guards

Before you start

1. Confirm that the machine is installed securely and is suitable for the fabric and seam type.
2. Inspect needles, loopers, presser foot, thread guides, and guards for damage.
3. Check thread type, needle size, and threading path against the machine instructions.
4. Verify oil level where applicable and check for leaks or unusual residue.
5. Ensure the needle area and table are clear of tools and loose materials.
6. Test the foot pedal, speed control, and safety devices before sewing.

How to use

1. Switch off the machine before threading, changing needles, or making adjustments.
2. Thread the machine through the specified guides and tension controls according to the threading diagram.
3. Fit the correct needle and confirm that it is seated and tightened properly.
4. Place the fabric under the presser foot with the seam aligned to the guide.
5. Lower the presser foot and start sewing at a low speed to verify stitch formation.
6. Increase speed gradually while guiding the fabric without pulling or pushing it forcefully.
7. Monitor stitch formation, thread tension, fabric feeding, and machine noise during operation.
8. Stop the machine if stitches skip, thread breaks repeatedly, or the fabric jams.
9. Raise the needle and presser foot as specified before removing the sewn material.
10. Switch off and isolate power before clearing a jam or accessing the needle and looper area.
11. Trim loose threads with suitable scissors and inspect the seam for uniformity.
12. Clean lint from accessible areas and follow the manufacturer's lubrication schedule.

⚠ Safety

- Only trained operators should use a high-speed interlock machine.
- Keep fingers away from the needle, loopers, and moving feed mechanism.
- Do not operate with missing guards, a bent needle, or damaged electrical controls.
- Switch off and isolate power before threading, needle replacement, jam clearing, or maintenance.
- Tie back long hair and avoid loose clothing, jewellery, or dangling accessories.
- Keep hands clear of the presser foot and do not leave the machine running unattended.

Care & storage

- Remove lint and thread fragments from the needle and looper area after use.
- Check needle condition and replace bent, blunt, or damaged needles.
- Maintain the lubrication system according to the manual.
- Inspect the belt, pedal, guards, and electrical cable regularly.
- Arrange periodic servicing to maintain stitch quality and safe operation.

Common mistakes to avoid

- Threading the machine incorrectly or using unsuitable thread and needle sizes.
- Pulling fabric forcefully and causing needle breakage or uneven stitches.
- Operating at high speed before confirming correct stitch formation.
- Clearing a thread jam while the machine is running.
- Allowing lint to accumulate around the loopers and feed mechanism.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.